

ECON1624

How AI and Digital Platforms are transforming

# TAXI SERVICE INDUSTRY

Presented by:



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# PRE-DIGITAL RIDE-HAILING INDUSTRY IN VIETNAM AND GLOBAL OVERVIEW

01

## Product

- **On-demand, point-to-point** rides mediated by the dispatcher
- Fixed metered fares

(Harding et al. 2016)

Information asymmetry presented in **principle-agent setting**

## INEFFICIENCIES

**Ex-ante adverse driver selection** risks as riders don't have complete information (hidden characteristics) (Harding et al. 2016)

**Ex-post moral hazard** as the riders cannot fully observe or verify the quality of the service (e.g., the length of taxi journey) and its price (hidden information) (Harding et al. 2016)

No digital receipt or post-ride accountability

## RESULT

Drivers' **opportunistic behaviour**: overcharge, take a circuitous route, or operate a substandard vehicle (hidden action) (Harding et al. 2016)

High transaction costs: negotiation costs (haggling over fares) and information costs (customers spend time verifying legitimacy) arise

**Trust deficit**: 34% complaining taxi meters and 12% deprecated the taxi driver's attitude (Quang 2021)

02

## Process

- **Offline booking** via telephone dispatch center or roadside hailing
- **Dispatch** by human with **no GPS precision**

(Khalfan et al. 2025; Chen and Lee 2020)

## INEFFICIENCIES

Without GPS-based data, the operators are unable to observe where demand and supply, thereby, allocating efficiently

Limited mechanism to **align driver supply** with **passenger demand**, creating **coordination failure**

Driver positioning by intuition: depends on the drivers' local knowledge

## RESULT

High transaction costs: monitoring costs (dispatch monitor fleet performance) and searching costs (customer locate available drivers)

Unnecessary traffic congestion caused by poor route optimization (Crammer and Kruger 2016)

High dispatch latency: high empty-miles rate due to idle drivers (Crammer and Kruger 2016)

E.g., in Los Angeles, taxi drivers travel 1.46 empty miles for every passenger mile, while UberX drivers travel only 0.56 empty miles per passenger mile.

# PRE-DIGITAL RIDE-HAILING INDUSTRY IN VIETNAM AND GLOBAL OVERVIEW

03

## Organization

- Dominated by **capital- intensive fleet operators** that either owned vehicles or leased them to drivers
- Operating a taxi required a government-issued licence.

E.g., Vinasun and Mai Linh operated extensive fleets with thousands of vehicles and a formal organizational structure (*Khalfan et al. 2025*)  
 ⇒ More care means more **competitive advantages**

## INEFFICIENCIES

- Government regulations cause a artificial or monopolistic rent through medallion prices for medallion owners
- Licensed holders prioritize extracting income rather than contributing to the service quality  
*(Cetin and Eryigit 2013)*
- High initial investment costs: fleet of vehicles, licenses, a brand, a call center, and operations management.

## RESULT

- High transaction costs: negotiation costs, searching/ information costs
- Entry barriers are higher for newcomers if they do not have capital to invest

04

## Market

- Fare schedules were standardized by regulatory bodies

*(Crammer and Kruger 2016)*

## INEFFICIENCIES

- Fixed tariffs could not reflect demand during the peak hours (*Crammer and Kruger 2016*)
- The market followed the **linear market's characteristics** rather than **multi-sided market**
- Department of Transport capped the number of licenses to prevent oversupply, creating a high entry barrier regardless of unmet passenger demand

## RESULT

- The incumbents were protected from competition by regulators
- No incentive to innovate  
*(Cetin and Eryigit 2013)*
- Supply shortage during peak hours as the price didn't reflected dynamically

# AI IN RIDE-HAILING

1

## AI applications in the industry

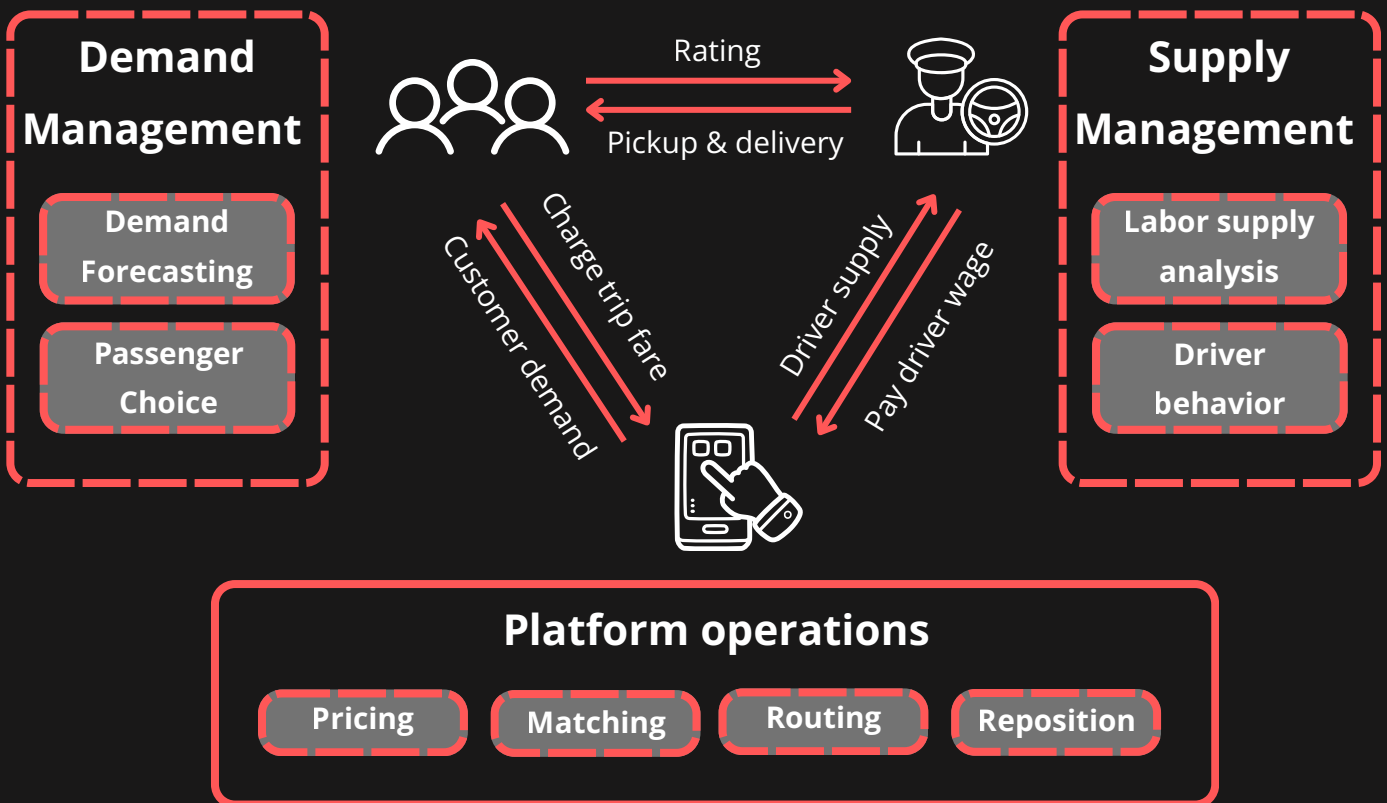


Figure 1: System overview of current on-demand ride-hailing service  
(Author) (Source: Liang et al. 2025)

»»» The current applications of AI within the ride-hailing industry signifying AI as a **predictive machine** that can help streamline the operations

### Details

## AI in Demand Management

### Demand Forecasting

Being used for **time-series analysis** to capture spatio-temporal patterns in data

»»» **Recurrent Neural Networks (RNNs)** and **Long Short-term Memory (LSTM)** networks are **supervised** machine learning models

(DigitalDefynd 2026; Liang et al. 2025)

» Uber use generated insight to plan driver incentive, adjust pricing strategy, allocate resources...

(Liang et al. 2025)

### Passenger Choice Analysis

**Understanding customer choice** thorough range of preferences and budgets

»»» **Random Forest (RF)** or **Gradient Boosting**, **supervised tree-based** machine learning models

(Liang et al. 2025)

» Uber use AI to identify passengers' mode choice in Express, Black, Carpool, and SUV

(Liang et al. 2025)

## AI in Supply Management

### Driver Behavior Analysis

Machine learning examines work participation, routing patterns, and multi-homing (switching between platforms).

»»» Utilizing logistic regression and Random Forest

(Liang et al. 2025)

# AI IN RIDE-HAILING

1

## AI applications in the industry

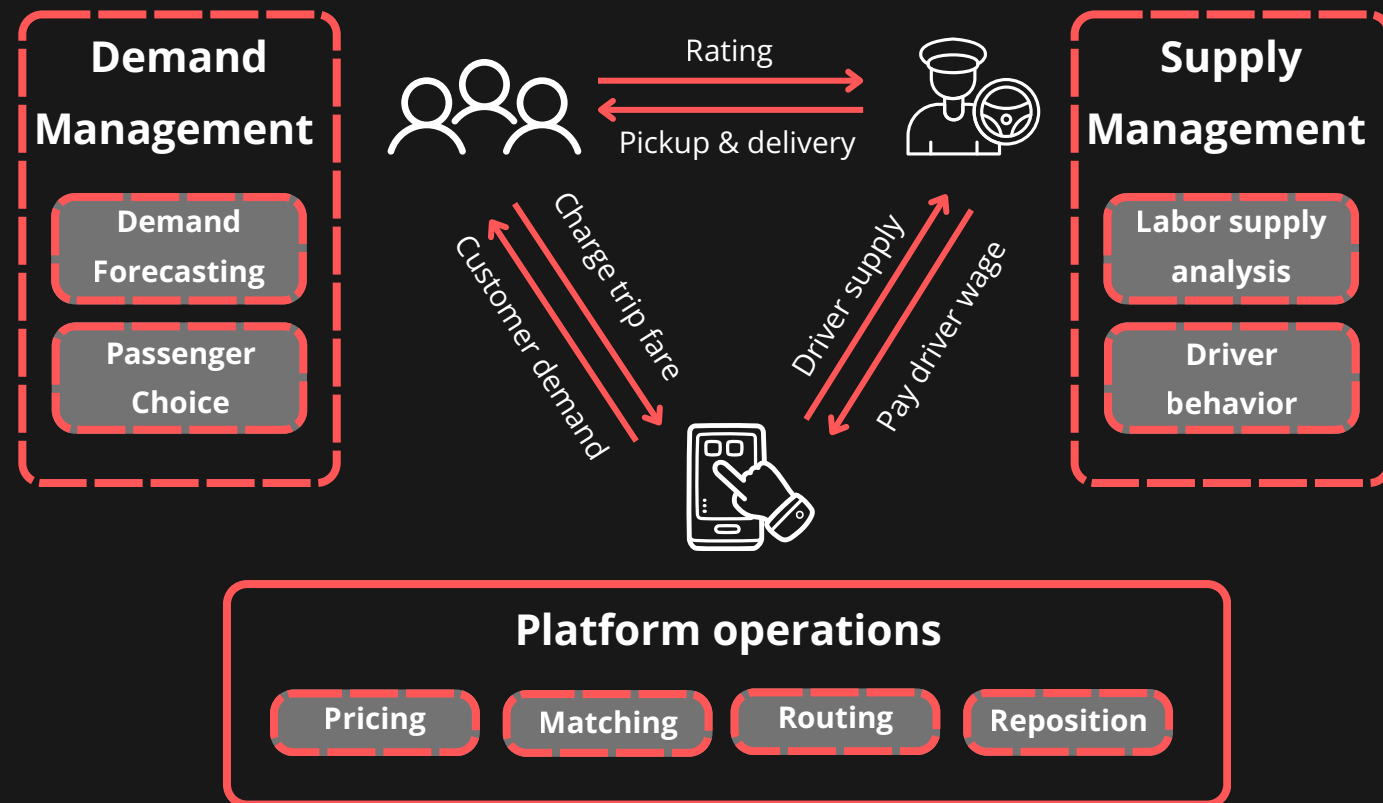


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### Details

## AI in Operations

### Pricing

**Pricing** must balance the interests of passengers and drivers.

»»» Reinforcement learning learns from historical demand, driver availability, and external factors to enable dynamic pricing

➤ **Reduce “wild goose chase”** inefficiency, leading to more **efficient use of drivers’ resources**

(Liang et al. 2025)

(DigitalDefynd 2026; Liang et al. 2025)

### Matching & Order dispatching

**Optimises driver-passenger pairing**, balancing short-term wait times against long-term driver earnings.

»»» Reinforcement Learning learn this as a sequential decision process

(Liang et al. 2025)

➤ Michelangelo, Uber’s reinforcement machine learning model, can generate more than 30M+ match pair predictions in real-time

(Uber 2017)

### Route Planning & ETA

Supervised learning (RNNs, Wide-Deep-Recurrent networks) **predicts the ETA and plans optimize route** using GPS trajectories and traffic data.

(Liang et al. 2025)

➤ Uber’s DeepETA, was trained on 1.4 billion ETA requests, serve prediction at median latency of 3.25ms

(Hu et al.2022)

### Vehicle Repositioning

Reinforcement Learning proactively **guides idle drivers** to **high-demand zones**.

»»» GNNs learn road-network representations to improve repositioning decisions.

(Liang et al. 2025)

➤ Grab’s drivers utilized GrabRideGuide, which leverage AI to analyze real-time data, reducing idle driving distance by 20%

(Vietdata 2025)

# AI IN RIDE-HAILING

2

## Creative Destruction caused by AI

### What is creative destruction?

the process by which new technologies and business models disrupt and eventually replace older ones.

(Alm and Cox n.d.)

| Dimension    | Pre-digital                                                                                                                                                                                                                                   | Post AI Implementation                                                                                                                                                                                                                                                                                                                                                                     |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product      | <ul style="list-style-type: none"><li>On-demand rides mediated by a human dispatcher</li><li>Fixed metered fares set by government tariffs</li><li>No advance booking, no driver visibility, no pre-trip fare estimate.</li></ul>             | <ul style="list-style-type: none"><li>On-demand rides with real-time driver tracking, upfront fare estimates, dynamic pricing, service tiers, and ETA eliminate information asymmetry.</li></ul>                                                                                                                                                                                           |
| Process      | <ul style="list-style-type: none"><li>Offline booking via telephone dispatch centres or roadside hailing.</li><li>Driver positioning by intuition (clustering at airports/hotels)</li><li>Cash-only payment. No GPS-based matching.</li></ul> | <ul style="list-style-type: none"><li>AI-powered algorithmic dispatch.</li><li>Eliminate fixed tariffs, enabling dynamic pricing that reflects the availability of driver supply and passenger demand.</li><li>Eliminate idle cruising by continuously matching drivers with passenger demand, increasing capacity utilization, where charge less but remaining the same revenue</li></ul> |
| Organization | <ul style="list-style-type: none"><li>Human dispatch workforce as core operational layer</li><li>Driver management based on manual oversight</li></ul>                                                                                        | <ul style="list-style-type: none"><li>Automatically coordinate drivers through algorithmic layer and demand forecasting</li><li>Driver behavior analysis enables data-driven labor data-driven driver supply management</li></ul>                                                                                                                                                          |
| Market       | <ul style="list-style-type: none"><li>Price is a fixed-contract variable set by the regulatory bodies</li></ul>                                                                                                                               | <ul style="list-style-type: none"><li>AI-enabled dynamic pricing restores the price signal as a market coordination mechanism</li></ul>                                                                                                                                                                                                                                                    |

(DigitalDefynd 2026; Khalfan et al. 2025; Liang et al. 2025; Vietdata 2025; ; Chen and Lee 2020; Harding et al. 2016; Cetin and Eryigit 2013)

# DIGITAL PLATFORMS AND MULTI-SIDED MARKET TRANSFORMATION

1

## The transformation

“Uber was founded as a response to the sustained failure of the taxi industry and its regulators to meet passenger demand”

Besides **push factors** being analyzed in the traditional industry, the **pull factors** also contribute largely to the transformation:

### Smartphone and GPS penetration

- With the increase in penetration rate of internet, mobile phone, and mobile internet, this improve digital connectivity, facilitating the adoption of app-based platform (Chalermpong et al. 2022)

### Rapid urbanization lead to rapid motorization

- The transportation infrastructure could not keep pace with urbanization in Southeast Asia, therefore; motorization is more preferred (Chalermpong et al. 2022)

### A large pool of private vehicle owners willing to supply rides

- informal transport operators (xe ôm in Vietnam) embraced platforms because it allowed them to access passengers without joining costly drivers' associations (Chalermpong et al. 2022)

# DIGITAL PLATFORMS AND MULTI-SIDED MARKET TRANSFORMATION

2

## Digital Platform Evolution

### PLATFORM

is an intermediary acts as a matching device, **facilitating interactions** between each side of the market to find the best agent on the other side and **generate the highest surplus**

(Jullien and Sand-Zantman 2021)



### MULTI-SIDED MARKET

A platform can tranform from two-ided to multi-sided market when it **connects more distinct user groups** whose value depends on the **presence of the others**

(Jullien and Sand-Zantman 2021)



### TRANSACTION COSTS

A platform with multi-sided externalities will further reduce transaction costs

## CASE STUDY: GRAB

| Year | Grab's Evolution                                | New Sides Added                              |
|------|-------------------------------------------------|----------------------------------------------|
| 2012 | GrabRides                                       | Drivers (private vehicle owners), riders     |
| 2016 | GrabPay, Grab Insurance, Grab Finance, GrabFood | FinTech users, food merchants, courier users |
| 2019 | GrabMerchant                                    | Businesses, advertisers                      |

(Ghazali 2025)

| Transaction costs      | How Platform resolves it?                                                                                                              |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Searching/ Information | Real-time GPS matching being integrated in platform, passenger sees the route, and upfront fares are displayed before making decision. |
| Bargaining             | Dynamic pricing signals market, upfront fares reduce post-trip disputes                                                                |
| Enforecement           | Bilateral rating, digital receipt being provided, in-app dispute resolution                                                            |

# DIGITAL PLATFORMS AND MULTI-SIDED MARKET TRANSFORMATION

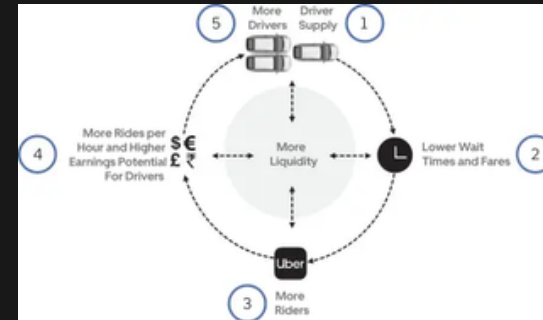
3

## Multi-side characteristic of ride-hailing platform

### CROSS-SIDE NETWORK EFFECT

- As more drivers join Uber, the platform will offer a **shorter waiting time**, which will attract riders
- As more riders use Uber, drivers can gain **additional value in more rides per hour** and a **higher earning potential**

» The increase in the number of people or activity in one group can benefit the other groups.



(Currier 2019)

### THE CHICKEN-AND-EGG PROBLEM

To attract new users, late comers must invest millions of dollars in the form of discount (for riders) and incentive (for drivers) to help them initially generate the network effect

» Didi allocated \$10 billion in subsidies to new users to compete with Uber in the Chinese ride-hailing market in 2015-2016

(Zhang et al. 2023; Ghasemi et al. 2023)

- Below-cost fares in growth phase for customers and then apply surge pricing as market matures
- Trigger surge pricing to increase fares to encourage more drivers to come online
- » Uber has capped surge pricing during crises and enhanced price transparency to both ensure driver availability for riders and driver's incentive

### PRICING STRATEGIES

(Lau 2025)

- Lock-in effects** existed within the ride-hailing app, including saved work addresses for quick access, linked payment methods, and exclusive deals, etc.,
- Due to network effect of existing market players, late comers are harder to participate in the market
- » Gojek cannot overcome accumulated switching costs and network effects of Grab and Xanh SM even though invested significant amount, making their market share drop from 30% to 7% and exited the market in mid-2024.

(Ghazali 2025; Ghasemi et al. 2023)

### WINNER-TAKES-ALL-DYNAMICS

# AI IN RIDE-HAILING

3

## Creative Destruction caused by platform

| Dimension    | Pre-digital                                                                                                                                                                                                                          | Post Platform Implementation                                                                                                                                                                                                                                                                                                   |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product      | <ul style="list-style-type: none"><li>Standardized metered ride</li><li>Limited differentiation between different ride-hailing service providers</li><li>No fare estimated</li></ul>                                                 | Dynamic surge pricing, upfront fares being shown on the app, super-app ecosystem (food, payments, insurance,...)                                                                                                                                                                                                               |
| Process      | <ul style="list-style-type: none"><li>Telephone dispatch</li><li>Cash-only payment</li><li>Limited information before, during, and after the rides</li></ul>                                                                         | <ul style="list-style-type: none"><li>Cashless digital payment (GrabPay, ZaloPay, MoMo) are enabled</li><li>Accountability being improved through bilateral ratings, digital receipts, and in-app dispute resolution</li><li>Every information of the rides are displayed in app with the driver details, fares, ...</li></ul> |
| Organization | <ul style="list-style-type: none"><li>Capital-intensive fleet operators (Vinasun, Mai Linh) with fleet and government-issued license</li></ul>                                                                                       | Asset-light pure platform (Grab, Be, or Uber) which they own no vehicles; however, they still have to invest in matching facilities and gain license to operate.                                                                                                                                                               |
| Market       | <ul style="list-style-type: none"><li>Linear one-sided market single driver-passenger transaction, limiting network effects</li><li>Fixed tariffs disabled the price mechanism; incumbents protected by regulatory capture</li></ul> | Multi-sided market with cross-side network effects: On Uber as well as other ride-hailing platform business, with more drivers, riders will have shorter waiting time and with more riders, drivers will have earning potential                                                                                                |

(Ghazali 2025; Lau 2025; Ghasemi et al. 2023; Zhang et al. 2023; Chalermpong et al. 2022; Currier 2019)

# RECOMMENDATIONS FOR VIETNAMESE SMES

1

## AI-powered solution propose

### Quality Management Issues

#### Poor Parts Quality

- Surface defects
- Failure to adhere to quality specifications
- Incorrect or substandard materials
- Wrong mix of parts



#### Poor Supplier Performance Quality

- Lack of effective communication
- Failure to satisfy to regulatory requirements
- Incorrect shipping paper work or packaging method

Suggestions

»»» **15-20%** of the revenue is the average losses caused by quality management issues, some could reach 40% (VPIC 2024)

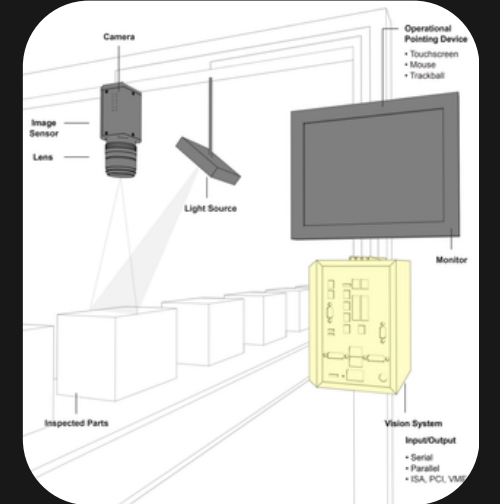
»»» Vietnamese SMEs in manufacturing need a proactive way to detect issues and ensure timely action.

## AI-powered + IoT for quality control

- With computer vision, AI enables real-time quality control and defect management
- IoT sensors will be deployed to capture the products' condition or performance in real-time

»»» Ensure quality is being controlled in real time

E.g., combining **deep-learning soft sensors** with a **high-resolution optical quality-control camera** improved industrial visual inspection in the printing sector, achieving an **automated classification rate** of **98.4%** of defect products.



### How should SMEs start using this technology?

- Data infrastructure is needed for training AI, as supervised machine learning will be used for classifying good and defective products, thereby, labelling AI is compulsory
- Requiring hardware prerequisites as IoT sensors and systems, require stable power, network connectivity, and infrastructure in the production line
- Employees' skills and competencies generated from a conducive company culture can enhance AI adoption

### System Risk Managing

- Ensure AI transparency and ethics are critical to prevent "black-box" problem, therefore, complying with ASEAN Guide on AI Governance and Ethics is recommended

(Dey et al. 2023; Nguyen and Tran 2023; Villalba-Diez et al.2019; ASEAN n.d.)

# RECOMMENDATIONS FOR VIETNAMESE SMES

## 2 Participating in platform market propose

### Drivers for plaform business adoption

- Necessity of establishing fluid partnership
- Adoption fatigue and adoption costs causing by adopting high technological complexities
- Perceiving the platform as guaranteeing great benefits for efficiency and security of sourcing.

»»» SMEs have high level of intentioni to participate in two-sided platform

(Bai et al. 2026; Marzi et al. 2023)

### Suggestions

### Participating in B2B platform market

- Facilitate market expansion, connecting SMEs to regional or even international buyers upon listing
- Operate within a flexible supply network where they can expand supplier selection, improve lead time, manage costs constraints...

»»» Met the SMEs constraints and needs

### How should SMEs start using this technology?

- Register on one relevant platform; invest in profile completeness and product specifications
- Quality factors needed to be emphasized since it's associated with SMEs' capacity to expand network through platform interaction
- Use platform analytics to identify high-performing products, geographies, and price points

### System Risk Managing

- lock-in risks arise when SMEs deepen their synergy with digital platforms; therefore, multi-homing is recommended

(Bai et al. 2026; Marzi et al. 2023)

**THANK YOU FOR  
LISTENING**



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